

## PAPER

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[View Journal](#) | [View Issue](#)Cite this: *J. Mater. Chem. A*, 2022, **10**, 15614Ammonium ion pre-intercalation stabilized tunnel  $h$ -WO<sub>3</sub> for fast NH<sub>4</sub><sup>+</sup> storage†Qiang Chen,<sup>a</sup> Mengda Song,<sup>a</sup> Xiangyong Zhang,<sup>b</sup> Jianli Zhang,<sup>a</sup> Guangya Hou<sup>a</sup> and Yiping Tang<sup>id</sup> <sup>\*a</sup>

The development of aqueous electrochemical energy storage devices based on ammonium ion (NH<sub>4</sub><sup>+</sup>) carriers has received increasing attention. However, the easy dissolution of electrode materials in an aqueous solution leads to the collapse of the host structure, which seriously hinders its application. Here, we report a pre-intercalated non-metal ion (NH<sub>4</sub><sup>+</sup>) stabilized tunnel  $h$ -WO<sub>3</sub> anode for fast NH<sub>4</sub><sup>+</sup> storage. The (NH<sub>4</sub>)<sub>x</sub>WO<sub>3</sub> anode and the  $\alpha$ -MnO<sub>2</sub> cathode with a tunnel structure constitute a novel ammonium-ion hybrid supercapacitor (A-HSC). The newly developed (NH<sub>4</sub>)<sub>x</sub>WO<sub>3</sub>/ $\alpha$ -MnO<sub>2</sub> A-HSC delivers an ultra-high areal capacitance of 2239.7 mF cm<sup>-2</sup> and a peak areal energy density of 1010.1  $\mu$ Wh cm<sup>-2</sup>. First-principles calculations show that ammonium ion pre-intercalated tunnel structured WO<sub>3</sub> can effectively enhance the NH<sub>4</sub><sup>+</sup> adsorption energy and reduce the NH<sub>4</sub><sup>+</sup> diffusion energy barrier. *Ex situ* characterization confirmed that the intercalation/deintercalation of NH<sub>4</sub><sup>+</sup> is accompanied by the formation/breaking of hydrogen bonds. These findings provide a new prototype for NH<sub>4</sub><sup>+</sup>-based energy storage and facilitate a basic understanding of the storage mechanism of NH<sub>4</sub><sup>+</sup>-tunnel structured metal oxide.

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## Introduction

Aqueous rechargeable batteries/supercapacitors are an important research direction for clean and sustainable energy development.<sup>1–3</sup> Although Li-ion batteries (LIBs) have shown great success in achieving high energy density ( $\sim 300$  Wh kg<sup>-1</sup>), issues such as safety and the scarcity of Li (20 ppm) seriously hinder their application.<sup>4,5</sup> Other metal ion (such as Na<sup>+</sup>,<sup>6,7</sup> K<sup>+</sup>,<sup>8</sup> Zn<sup>2+</sup>,<sup>9</sup> Mg<sup>2+</sup>,<sup>10</sup> Ca<sup>2+</sup>,<sup>11</sup> Al<sup>3+</sup>,<sup>12</sup> etc.) and non-metal cation (such as H<sup>+</sup><sup>13</sup> and NH<sub>4</sub><sup>+</sup><sup>14</sup>) charge carriers in aqueous batteries/supercapacitors, are an important alternative to traditional LIBs. Among them, aqueous NH<sub>4</sub><sup>+</sup> energy storage systems have become competitive candidates due to their intrinsic safety and abundant charge carriers (which can be synthesized from H and N sources).<sup>15,16</sup> Compared with the relatively cheap Na<sup>+</sup> and Li<sup>+</sup>, NH<sub>4</sub><sup>+</sup> shows a lighter molar mass ( $\sim 18$  g mol<sup>-1</sup>, 23 g mol<sup>-1</sup> for Na<sup>+</sup>) and smaller hydrated ionic radius (3.31 Å, 3.38 Å for Na<sup>+</sup>), which leads to a faster diffusion rate of NH<sub>4</sub><sup>+</sup> in the aqueous electrolyte, which is beneficial in improving the power density of the device.<sup>17,18</sup> Compared with spherical metal ions without preferential orientation, NH<sub>4</sub><sup>+</sup> with a tetrahedral structure displays rich topological chemistry in host materials,

can be intercalated into electrodes by forming covalent-ionic bonds, and can also act as a reversible redox center for charge transfer.<sup>19,20</sup> During the intercalation/de-intercalation process, the charge transfer between NH<sub>4</sub><sup>+</sup> and the electrode can be carried out through the formation/breaking of hydrogen bonds, so as to achieve a higher energy storage capacity than metal ions, which is more advantageous in terms of energy density.<sup>21</sup> Importantly, an NH<sub>4</sub><sup>+</sup> energy storage system has significant advantages in recycling compared with traditional metal-ion batteries due to the characteristics of ammonium salts that are easily soluble in water and easily decomposed by heat.<sup>22</sup> In addition, NH<sub>4</sub><sup>+</sup> is less corrosive and has less polarized hydrogen evolution than H<sup>+</sup> or H<sub>3</sub>O<sup>+</sup>.<sup>23</sup> Therefore, the development of NH<sub>4</sub><sup>+</sup> storage devices has attracted great attention.

To date, significant research progress has been made on NH<sub>4</sub><sup>+</sup>-based aqueous energy storage devices, and a series of electrode materials suitable for NH<sub>4</sub><sup>+</sup> storage have been developed, mainly based on the following two ideas. (i) Using coordination compounds with 3D frameworks and large interstitial sites for NH<sub>4</sub><sup>+</sup> storage, such as Prussian blue analogues ((NH<sub>4</sub>)<sub>1.47</sub>Ni[Fe(CN)<sub>6</sub>]<sub>0.88</sub>,<sup>18</sup> Fe<sub>4</sub>[Fe(CN)<sub>6</sub>]<sub>3</sub>,<sup>24,25</sup> NaFe<sup>III</sup>Fe<sup>II</sup>(CN)<sub>6</sub>,<sup>26</sup> and copper hexacyanoferrate<sup>27</sup>). (ii) Using layered metal oxide electrodes with large interlayer sites to achieve high capacity through the interlayer topological chemistry of NH<sub>4</sub><sup>+</sup>, such as Ti<sub>3</sub>C<sub>2</sub> MXenes,<sup>28</sup> metal oxides (V<sub>2</sub>O<sub>5</sub>,<sup>21</sup> NH<sub>4</sub>V<sub>4</sub>O<sub>10</sub>,<sup>29</sup> MnO<sub>x</sub>,<sup>20</sup> and MoO<sub>3</sub><sup>30</sup>), etc. Recently, the unique properties of tunnel-structured metal oxide materials with large storage channels in electrochemical energy storage have attracted extensive

<sup>a</sup>College of Material Science and Engineering, Zhejiang University of Technology, Hangzhou 310014, China. E-mail: tangyiping@zjut.edu.cn<sup>b</sup>College of Material Science and Engineering, Shenzhen University, Shenzhen 30035, China† Electronic supplementary information (ESI) available. See <https://doi.org/10.1039/d2ta03643f>

attention of researchers. Zhi's group reported the tunnel structure of the *h*-WO<sub>3</sub> anode and proposed that there is reversible NH<sub>4</sub><sup>+</sup> intercalation/de-intercalation and formation/fracture of H bonds in the tunnel structure, and it exhibits fast electrode dynamics.<sup>30</sup> Zhang *et al.* also confirmed that tunnel *h*-WO<sub>3</sub> has a better NH<sub>4</sub><sup>+</sup> storage capacity than *m*-WO<sub>3</sub>.<sup>31</sup> Although the above studies have preliminarily confirmed that metal oxides with tunnel structure have excellent NH<sub>4</sub><sup>+</sup> storage performance. However, it has poor structural stability during charge storage, and is prone to dissolve in (weakly) acidic aqueous solutions, resulting in abnormal structural collapse.<sup>32</sup> Therefore, it is of great significance to explore a new stabilization strategy for tunnel-structured electrodes to realize synergistic improvement of the stability and capacity of tunnel-structured NH<sub>4</sub><sup>+</sup> storage devices.

Herein, we report a pre-intercalated non-metal ion (NH<sub>4</sub><sup>+</sup>) stabilized tunnel *h*-WO<sub>3</sub> anode for fast NH<sub>4</sub><sup>+</sup> storage, and construct an ammonium ion hybrid supercapacitor (A-HSC) based on the tunnel-structured (NH<sub>4</sub>)<sub>x</sub>WO<sub>3</sub> anode and  $\alpha$ -MnO<sub>2</sub> cathode. The (NH<sub>4</sub>)<sub>x</sub>WO<sub>3</sub> nanorods grown on flexible carbon cloth are prepared by a one-step hydrothermal process. In an A-HSC, intercalation/de-intercalation and hydrogen bonding formation/fracture of the tunnel structured (NH<sub>4</sub>)<sub>x</sub>WO<sub>3</sub> nanorod anode achieve outstanding electrochemical performance. The prepared (NH<sub>4</sub>)<sub>x</sub>WO<sub>3</sub>/MnO<sub>2</sub> A-HSCs exhibit an ultrahigh areal capacitance of 2239.7 mF cm<sup>-2</sup> at 2 mA cm<sup>-2</sup>, and a wide operating voltage window of 0–1.8 V. They also delivered an outstanding areal energy density of 1010.1  $\mu$ Wh cm<sup>-2</sup> and a peak areal power density of 18.0 mW cm<sup>-2</sup>. In addition, the rechargeable aqueous A-HSC showed an extraordinary capacity

retention rate of 78.6% after 13 600 cycles. Impressively, the results of the experiments and first-principles calculations indicate that NH<sub>4</sub><sup>+</sup> pre-intercalation can significantly optimize the WO<sub>3</sub> tunnel structure environment. Furthermore, NH<sub>4</sub><sup>+</sup> intercalates and diffuses into the (NH<sub>4</sub>)<sub>x</sub>WO<sub>3</sub> tunnel structure through the H bond formation/fracture during the charge-discharge process, leading to higher adsorption energy and ultrafast reaction kinetics.

## Results and discussion

To explore the NH<sub>4</sub><sup>+</sup> storage properties of non-metal ion pre-intercalated tunnel-structure WO<sub>3</sub>, we carried out a systematic first-principles calculation. We first calculated the adsorption energies of single NH<sub>4</sub><sup>+</sup>, an H<sub>2</sub>O and an NH<sub>4</sub><sup>+</sup>, and two NH<sub>4</sub><sup>+</sup> intercalations in the tunnel structured WO<sub>3</sub> (Fig. 1a). Interestingly, the inserted NH<sub>4</sub><sup>+</sup> not only forms an H bond with the adjacent O of WO<sub>3</sub>, but also forms an H bond with H<sub>2</sub>O. The length of the formed H bond is in the range of 1.76 to 2.07 Å (Fig. S1 and Table S1, ESI†). The calculation results show that intercalation of single NH<sub>4</sub><sup>+</sup> in the WO<sub>3</sub> tunnel structure has a strong adsorption of –2.09 eV. Pre-intercalation of single H<sub>2</sub>O in the tunnel structure will increase the adsorption of NH<sub>4</sub><sup>+</sup> to –2.28 eV, and pre-intercalation of NH<sub>4</sub><sup>+</sup> will further enhance the adsorption of NH<sub>4</sub><sup>+</sup> to an astonishing –3.78 eV. This indicates that the pre-intercalation of NH<sub>4</sub><sup>+</sup> can significantly improve the adsorption capacity of the tunnel structure for NH<sub>4</sub><sup>+</sup>, which is beneficial to the excellent NH<sub>4</sub><sup>+</sup> storage performance of (NH<sub>4</sub>)<sub>x</sub>WO<sub>3</sub>. To better understand the interaction mechanism between NH<sub>4</sub><sup>+</sup> and the tunnel structured WO<sub>3</sub>, we conducted

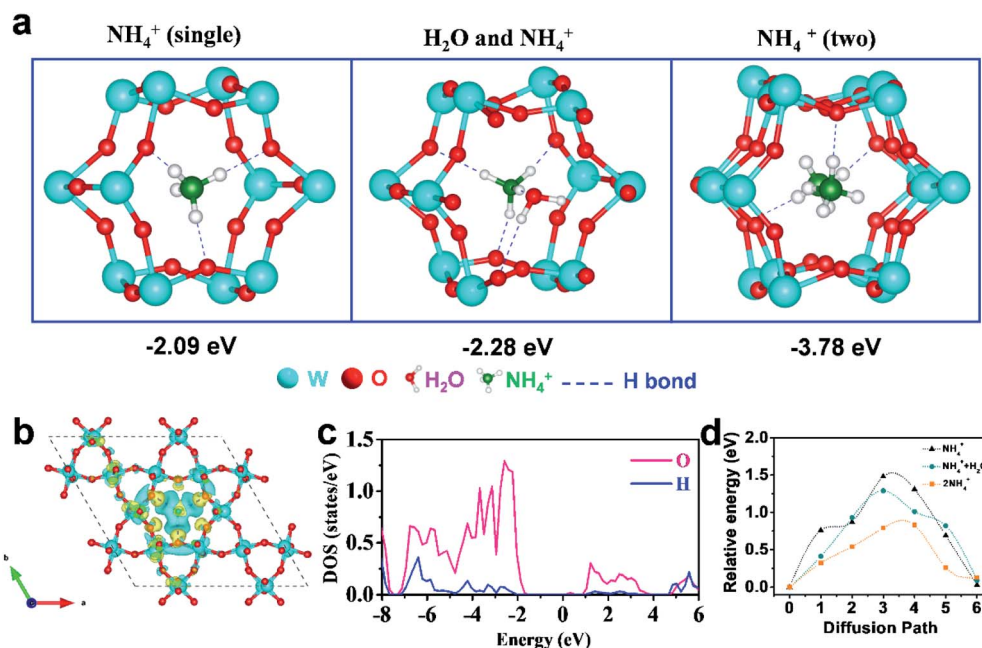


Fig. 1 (a) Schematic diagram of the three bonding states and adsorption energy in the WO<sub>3</sub> tunnel structure: single NH<sub>4</sub><sup>+</sup>, an H<sub>2</sub>O and an NH<sub>4</sub><sup>+</sup>, and two NH<sub>4</sub><sup>+</sup> intercalation. (b) Diagram of the charge density difference of NH<sub>4</sub><sup>+</sup> inserted tunnel *h*-WO<sub>3</sub>, where the charge densities of the isosurfaces are 0.001 eÅ<sup>-3</sup> for blue and 0.001 eÅ<sup>-3</sup> for yellow. (c) The PDOS of hydrogen and oxygen that form hydrogen bonds. (d) Diffusion activation energy of NH<sub>4</sub><sup>+</sup> along the WO<sub>3</sub> tunnel.

a differential charge density analysis. As shown in Fig. 1b, during the process of interaction between  $\text{NH}_4^+$  and the tunnel structured  $\text{WO}_3$ , electrons will be enriched with oxygen atoms in  $\text{WO}_3$  which is hydrogen-bonded with  $\text{NH}_4^+$ . This process will enhance the electronegativity of oxygen, thereby enhancing the hydrogen bond interaction between  $\text{NH}_4^+$  and  $\text{WO}_3$ . The strong bonding of hydrogen bonds makes tunnel  $\text{WO}_3$  more stable. Furthermore, we further studied the PDOS of hydrogen and oxygen that form hydrogen bonds. Fig. 1c shows that the H 1s and the O 2p orbitals overlap obviously, which further explains the strong interaction between  $\text{NH}_4^+$  and  $\text{WO}_3$ .

Finally, we systematically calculated the diffusion of  $\text{NH}_4^+$  in the tunnel structured  $\text{WO}_3$  (Fig. 1d). The diffusion energy barrier of a single  $\text{NH}_4^+$  is 1.48 eV. When the water molecules are pre-embedded, the diffusion energy barrier is reduced to 1.29 eV. Interestingly, when  $\text{NH}_4^+$  is pre-intercalated in the tunnel structure, the diffusion energy barrier will be greatly reduced, down to 0.83 eV. In summary, the pre-intercalation of  $\text{NH}_4^+$  in  $\text{WO}_3$  will not only enhance the host's adsorption energy for  $\text{NH}_4^+$ , but also reduce the diffusion energy of  $\text{NH}_4^+$ , thus playing an extremely critical role in the hydrogen bond network in the entire process. This result indicates that  $\text{NH}_4^+$  pre-intercalation in tunnel  $\text{WO}_3$  can achieve fast and stable storage of  $\text{NH}_4^+$ .

The schematic diagram of the synthesis process of  $(\text{NH}_4)_x\text{WO}_3$  is shown in Fig. 2a. The  $h\text{-WO}_3$  nanowires and  $\text{NH}_4^+$  stabilized tunnel  $h\text{-WO}_3$  ( $(\text{NH}_4)_x\text{WO}_3$ ) nanorods were synthesized by a simple one-step hydrothermal method. Interestingly,

when no ammonium source was added to the precursor solution, orthogonal structured  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$  was obtained (Fig. S2†). The crystal structure diagram of  $h\text{-WO}_3$  shows that the size of the tunnel structure is about 5.4 Å larger than the ionic radius of  $\text{NH}_4^+$  ( $\sim 1.48$  Å), confirming that  $h\text{-WO}_3$  is a very potential carrier for  $\text{NH}_4^+$  storage. The X-ray diffraction (XRD, Fig. 2b) patterns and Raman spectra were employed to investigate the phase composition of the samples. The XRD pattern of  $h\text{-WO}_3$  and  $(\text{NH}_4)_x\text{WO}_3$  displays a series of characteristic peaks at 13.9, 23.6, 28.1, and 36.8°, corresponding to the (100), (002), (200), and (202) diffractions of  $h\text{-WO}_3$  (JCPDS no. 85-2460) or  $(\text{NH}_4)_{0.25}\text{WO}_3$  (JCPDS no. 73-1084),<sup>33</sup> respectively. Comparatively,  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$  exhibits five diffraction peaks at 14.1, 18.1, 22.9, 28.1, and 36.6°, indexed to the (200), (111), (020), (202), and (222) diffractions of  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$  (JCPDS no. 87-1203),<sup>34</sup> respectively. The Raman spectra (Fig. S3†) of  $(\text{NH}_4)_x\text{WO}_3$  shows peaks at 277.7, 771.4, 917.3, and 957.7  $\text{cm}^{-1}$ , which are characteristic of  $h\text{-WO}_3$ .<sup>35</sup> The Raman (Fig. S3†) bands at 340.3, 675.6, 807.2, and 950.7  $\text{cm}^{-1}$  stand for  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$ .<sup>34</sup> Both XRD and Raman results revealed that the addition of an ammonium source in the precursor solution resulted in a significant change in the phase structure.

The scanning electron microscope (SEM) images in Fig. 2c and d show that the  $(\text{NH}_4)_x\text{WO}_3$  nanorods are uniformly covered on the surfaces of the CC. Interestingly, the cross-section of  $(\text{NH}_4)_x\text{WO}_3$  nanorods exhibits a hexagonal morphology, which matches with its hexagonal phase structure. SEM element mapping (Fig. S4†) shows that nitrogen, tungsten, and oxygen

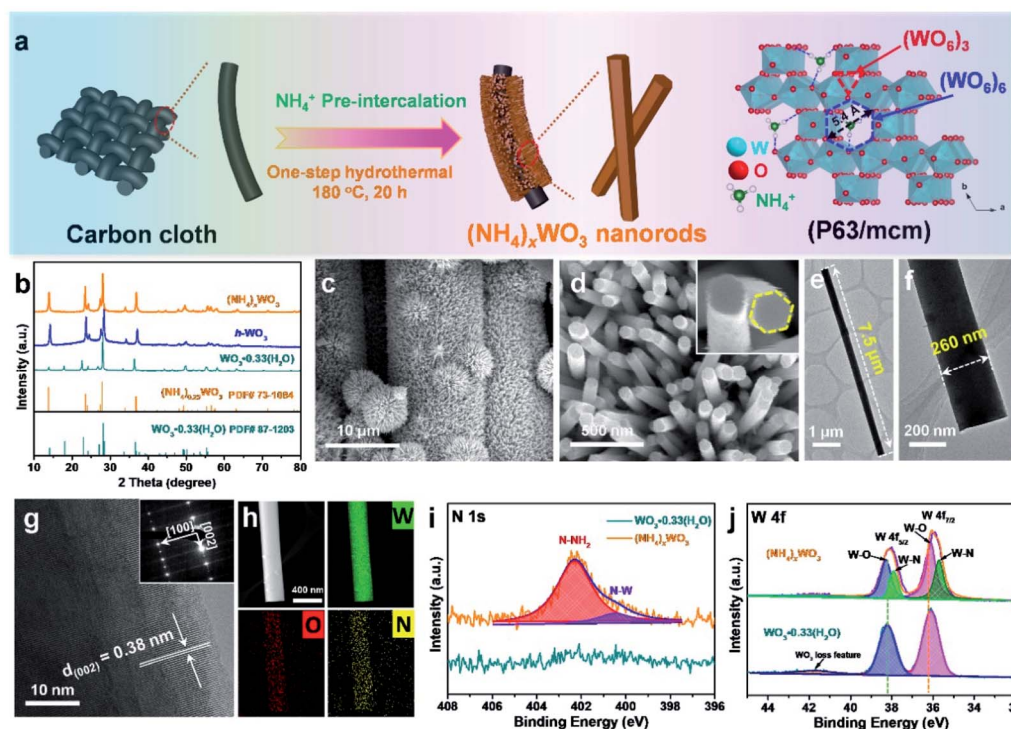


Fig. 2 Structural characteristics of the tunnel-structured  $(\text{NH}_4)_x\text{WO}_3$  anode. (a) Schematic diagrams of the structure and synthesis process of  $(\text{NH}_4)_x\text{WO}_3$ . (b) XRD patterns of  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$ ,  $h\text{-WO}_3$ , and  $(\text{NH}_4)_x\text{WO}_3$ . (c and d) SEM images, (e and f) TEM, (g) HRTEM, SAED (the inset), and (h) element mapping of  $(\text{NH}_4)_x\text{WO}_3$ . (i) Core-level N 1s and (j) W 4f XPS spectra of  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$  and  $(\text{NH}_4)_x\text{WO}_3$ .

are uniformly distributed on the surface of the samples, which confirms the composition of  $(\text{NH}_4)_x\text{WO}_3$ . Energy dispersive spectroscopy (EDS, Fig. S5†) analysis reveals a N/W molar ratio of  $\sim 0.29 : 1$ , agreeing well with the formula  $(\text{NH}_4)_{0.25}\text{WO}_3$ . The morphology of  $h\text{-WO}_3$  is nanowire layered balls grown on carbon cloth (Fig. S6†). Fig. S7† shows that the bulky morphology of  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$  on the CC surface is completely different from that of  $(\text{NH}_4)_x\text{WO}_3$  nanorods, which indicates that the introduction of an ammonium source promotes the transformation of the morphology. We further investigated the morphological features of  $(\text{NH}_4)_x\text{WO}_3$  by using the transmission electron microscope image (TEM, Fig. 2e and f), where the length and diameter of the nanorod are approximately  $7.5 \mu\text{m}$  and  $260 \text{ nm}$ , respectively.

The high-resolution TEM (HRTEM) image of  $(\text{NH}_4)_x\text{WO}_3$  shows that the neighboring lattice spacing along the radial direction is  $0.38 \text{ nm}$ , which is assigned to the feature of (002) planes (Fig. 2g). The selected area electron diffraction (SAED, inset of Fig. 2g) pattern ascertains that the  $(\text{NH}_4)_x\text{WO}_3$  nanorod has a single crystal nature. To reveal the element distribution along a single nanorod, the element mappings of N, W, and O are shown in Fig. 2h, in which the uniform coverage of nitrogen may be due to  $\text{NH}_4^+$  intercalation in the  $h\text{-WO}_3$  tunnel structure. For comparison, TEM, HRTEM, SAED, and element mapping of  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$  are provided in Figs S8 and S9†.

Moreover, X-ray photoelectron spectroscopy (XPS) was further used to determine the surface composition and chemical state of  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$  and  $(\text{NH}_4)_x\text{WO}_3$ . The W, O, and C

elements were detected in both samples while N was only detected in  $(\text{NH}_4)_x\text{WO}_3$  nanorods (Fig. S10†). The N 1s spectra of  $(\text{NH}_4)_x\text{WO}_3$  exhibits two peaks at  $400.5$  and  $402.3 \text{ eV}$ , corresponding to the bonding of N-W and N- $\text{NH}_2$ ,<sup>36</sup> revealing the composition of  $\text{NH}_4^+$  in  $(\text{NH}_4)_x\text{WO}_3$  (Fig. 2i). The W 4f XPS spectra of  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$  and  $(\text{NH}_4)_x\text{WO}_3$  show 4f 7/2 ( $36.1 \text{ eV}$ ) and 4f 5/2 ( $38.3 \text{ eV}$ ) peaks of W-O binding (Fig. 2j).<sup>37</sup> Compared with  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$ ,  $(\text{NH}_4)_x\text{WO}_3$  sample has additional W-N ( $35.7$  and  $37.9 \text{ eV}$ )<sup>38,39</sup> composition, again confirming the  $(\text{NH}_4)_x\text{WO}_3$  intercalation by  $\text{NH}_4^+$ . Based on the above results, the coordination structure of W and N in  $(\text{NH}_4)_x\text{WO}_3$  after  $\text{NH}_4^+$  intercalation is determined, which promotes the improvement of the stability and electrochemical performance of tunnel structured  $(\text{NH}_4)_x\text{WO}_3$ .

The electrochemical performance of  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$ ,  $h\text{-WO}_3$ , and  $(\text{NH}_4)_x\text{WO}_3$  electrodes was measured in a three-electrode configuration in  $2.0 \text{ M NH}_4\text{Ac}$ . The galvanostatic charge/discharge (GCD) profiles of  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$ ,  $h\text{-WO}_3$ , and  $(\text{NH}_4)_x\text{WO}_3$  are shown in Fig. 3a.  $(\text{NH}_4)_x\text{WO}_3$  demonstrates a much longer discharge time than  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$  and  $h\text{-WO}_3$ , suggesting its much higher capacitance. The CV curves (Fig. 3b) further confirm the higher specific capacitance of  $(\text{NH}_4)_x\text{WO}_3$ , as it shows a larger area than  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$  and  $h\text{-WO}_3$ . To further explore the rate performance of  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$ ,  $h\text{-WO}_3$ , and  $(\text{NH}_4)_x\text{WO}_3$ , GCD curves at various current densities are collected (Fig. S11 and S13†).  $(\text{NH}_4)_x\text{WO}_3$  achieves a remarkable areal capacitance of  $8.0 \text{ F cm}^{-2}$  at  $2 \text{ mA cm}^{-2}$  (Fig. 3c), which is more than twice that of  $h\text{-WO}_3$  ( $3.7 \text{ F}$

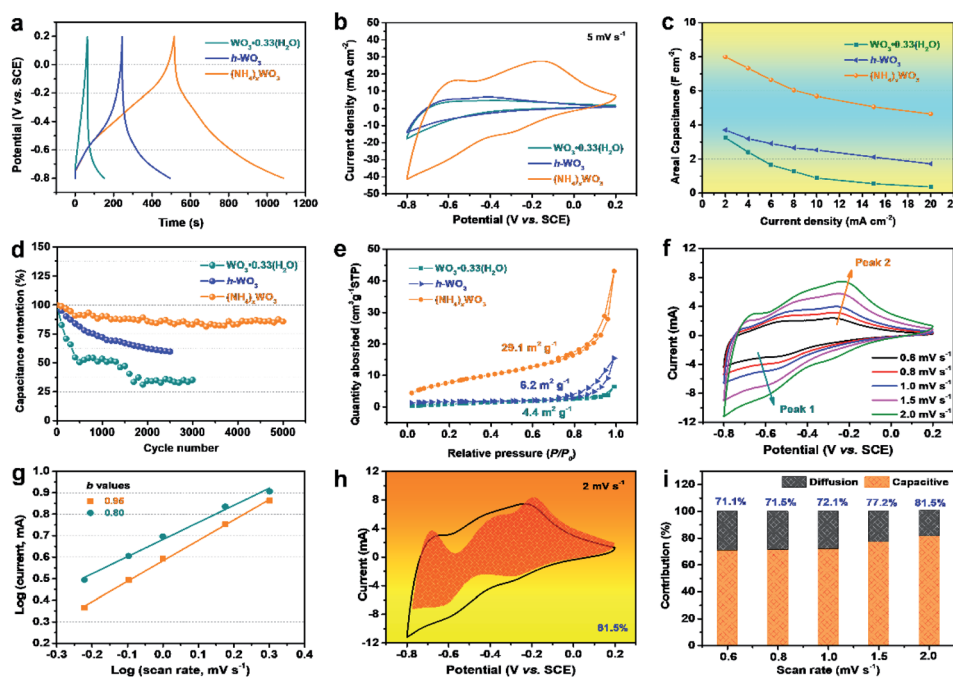


Fig. 3 Electrochemical performance of the anode. Comparison of (a) GCD profiles ( $10 \text{ mA cm}^{-2}$ ), (b) CV curves ( $5 \text{ mV s}^{-1}$ ), (c) areal capacitance as a function of the current densities, (d) cycling performance (at  $20 \text{ mA cm}^{-2}$ ), and (e) nitrogen adsorption isotherms (at  $77 \text{ K}$ ) of  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$ ,  $h\text{-WO}_3$ , and  $(\text{NH}_4)_x\text{WO}_3$ . (f) CV curves at scan rates of  $0.6$ – $2.0 \text{ mV s}^{-1}$ , (g) logarithm peak current versus logarithm scan rate plot, (h) pseudo-capacitance contribution at  $2 \text{ mV s}^{-1}$ , and (i) contribution ratio of the pseudo-capacitance and diffusion-controlled charge versus scan rates for the  $(\text{NH}_4)_x\text{WO}_3$  electrode.

$\text{cm}^{-2}$ ) and  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$  ( $3.3 \text{ F cm}^{-2}$ ). Such a superior areal capacitance is higher than the most reported values of HSC electrodes (Table S2†), such as  $\text{Na}_{0.5}\text{MnO}_2$  ( $0.37 \text{ F cm}^{-2}$  at  $1 \text{ mA cm}^{-2}$ ),<sup>40</sup>  $\text{Zn}_x\text{MnO}_2$  ( $1.75 \text{ F cm}^{-2}$  at  $2 \text{ mA cm}^{-2}$ ),<sup>41</sup> phosphate ion functionalized  $\text{Co}_3\text{O}_4$  ( $0.96 \text{ F cm}^{-2}$  at  $2.52 \text{ mA cm}^{-2}$ ),<sup>42</sup>  $\text{Co-Ni}_3\text{S}_2$  ( $3.6 \text{ F cm}^{-2}$  at  $8 \text{ mA cm}^{-2}$ ),<sup>43</sup>  $\text{W}_3\text{S}_2$  ( $0.93 \text{ F cm}^{-2}$  at  $4 \text{ mA cm}^{-2}$ ),<sup>44</sup> and  $\text{WO}_{3-x}$  ( $1.83 \text{ F cm}^{-2}$  at  $1 \text{ mA cm}^{-2}$ ).<sup>45</sup> In addition, long cycle stability is an important parameter to judge the stability of the electrode structure. As shown in Fig. 3d,  $(\text{NH}_4)_x\text{WO}_3$  exhibits excellent cycle performance with 85.8% retention of the initial capacity after 5000 cycles. In contrast,  $h\text{-WO}_3$  only provides 59.7% capacity retention after 2500 cycles, and  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$  only provides 35.2% capacity retention after 3000 cycles. This indicates that the  $\text{NH}_4^+$  pre-intercalated tunnel  $\text{WO}_3$  can further optimize structural stability. In addition, the Brunauer Emmett Teller (BET, Fig. 3e) surface area of the  $(\text{NH}_4)_x\text{WO}_3$  electrode ( $28.1 \text{ m}^2 \text{ g}^{-1}$ ) is significantly larger than that of  $h\text{-WO}_3$  ( $6.2 \text{ m}^2 \text{ g}^{-1}$ ) and  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$  ( $4.4 \text{ m}^2 \text{ g}^{-1}$ ). Therefore, the outstanding capacitance performance of  $(\text{NH}_4)_x\text{WO}_3$  is attributed to the enhanced electrochemical surface area.

To optimize the  $\text{NH}_4^+$  pre-intercalation content, the feed amount of  $(\text{NH}_4)_2\text{SO}_4$  in the precursor solution was varied from 0 to 6 g. The morphology of the prepared  $(\text{NH}_4)_x\text{WO}_3$  is highly correlated with the feed amount of  $(\text{NH}_4)_2\text{SO}_4$ . With an increase in the feed amount, the morphology changes from a block to nanorods, and the size of the nanorods gradually becomes smaller (Fig. S14†). Moreover, the phase structure also changed significantly after the addition of  $(\text{NH}_4)_2\text{SO}_4$  (Fig. S15†). Specifically, after adding 2.0 g of  $(\text{NH}_4)_2\text{SO}_4$  to the precursor solution, the phase changes from orthogonal  $\text{WO}_3 \cdot 0.33(\text{H}_2\text{O})$  to hexagonal  $(\text{NH}_4)_x\text{WO}_3$ , and the  $(\text{NH}_4)_x\text{WO}_3$  phase is still maintained by further increasing the addition amount (2.0–8.0 g).  $(\text{NH}_4)_x\text{WO}_3$  exhibits the highest capacitance when the addition amount of  $(\text{NH}_4)_2\text{SO}_4$  is 4.0 g (Fig. S16†). Therefore, we believe that 4.0 g is the optimal addition amount of  $(\text{NH}_4)_2\text{SO}_4$ .

To further study the electrochemical kinetics of the  $(\text{NH}_4)_x\text{WO}_3$  anode, CV curves were collected at various scan rates of 0.6–2.0  $\text{mV s}^{-1}$  (Fig. 3f). The two relatively symmetric CV peaks are related to the “chemisorption” of ions on the electrode surface during the redox reaction. This phenomenon indicates that the  $\text{NH}_4^+$ -tunnel  $(\text{NH}_4)_x\text{WO}_3$  system is not only a chemical system of intercalation/de-intercalation, but the formation of chemical bonds between  $\text{NH}_4^+$  and the host may play a key role in the CV behaviour. In general, the relationship between the scan rate ( $n$ ) and peak current ( $i$ ) is as follows:

$$i = av^b \quad (1)$$

$$\log(i) = b \log(n) + \log(a) \quad (2)$$

where  $i$  and  $n$  represent the current density and scan rate, respectively. Coefficients  $a$  and exponent  $b$  are fitting parameters. The calculation of the  $b$  value is expressed by using the slope of  $\log(n)$  and  $\log(i)$ , which can directly determine the reaction mechanism and electrochemical behaviour of  $\text{NH}_4^+$  storage. As shown in Fig. 3g, the scan rate and peak current show a good linear relationship. The  $b$  values are 0.95 and 0.8,

respectively, indicating that the  $(\text{NH}_4)_x\text{WO}_3$  electrode is mainly controlled by the surface reaction characteristics leading to excellent rate performance. To quantify the diffusion control and capacitance characteristics at different scan rates, we further calculated the pseudo-capacitance contribution. Fig. 3h shows that the pseudo-capacitance contribution of  $(\text{NH}_4)_x\text{WO}_3$  reaches 81.5% at  $2.0 \text{ mV s}^{-1}$ . This non-diffusion-controlled redox process is defined as intercalation pseudo-capacitance. In addition, the pseudo-capacitance contribution gradually increases as the scan rate increases, and the pseudo-capacitance contribution can reach 81.5% even at  $2 \text{ mV s}^{-1}$  (Fig. 3i and S17†). Therefore, kinetic analysis and capacitance contribution confirm that the charge storage of the  $(\text{NH}_4)_x\text{WO}_3$  electrode is mainly due to the pseudo-capacitance behaviour and the chemical bond between  $\text{NH}_4^+$  and the host, which is conducive to fast  $\text{NH}_4^+$  storage and long cycling.

$\alpha\text{-MnO}_2$  has a  $2 \times 2$  tunnel structure ( $\sim 4.76 \text{ \AA}$ ) connected by  $\text{MnO}_6$  octahedra sharing edges and vertices (Fig. S18a†), which can be used as the cathode to provide a suitable channel for storing  $\text{NH}_4^+$  (cation radius  $\sim 1.48 \text{ \AA}$ ). To evaluate the possibility of  $\text{NH}_4^+$  storage of tunnel structured  $\alpha\text{-MnO}_2$ , we performed systematic first-principles calculations. The insertion of a single  $\text{NH}_4^+$  can form an H bond with the adjacent O of the tunnel structured  $\text{MnO}_2$ , which has a strong adsorption of  $-1.97 \text{ eV}$  (Fig. S18b†). This indicates that the tunnel structured  $\alpha\text{-MnO}_2$  has excellent  $\text{NH}_4^+$  adsorption capacity, which also confirms the potential  $\text{NH}_4^+$  storage performance of  $\alpha\text{-MnO}_2$ . To reveal the interaction between  $\text{NH}_4^+$  and the tunnel structured  $\text{MnO}_2$ , a differential charge density analysis was performed (Fig. S18c†). When  $\text{NH}_4^+$  interacts with the tunnel structured  $\text{MnO}_2$ , the oxygen atoms in  $\text{MnO}_2$  that form hydrogen bonds with  $\text{NH}_4^+$  will be enriched in electrons. In addition, the projected density of states (PDOS) diagram of hydrogen and oxygen forming hydrogen bonds (Fig. S18d†) shows that the H 1s and O 2p orbitals overlap significantly, which further explains the strong interaction between  $\text{NH}_4^+$  and  $\text{MnO}_2$ . The above results all indicate that the tunnel structured  $\alpha\text{-MnO}_2$  has excellent  $\text{NH}_4^+$  storage capacity.

Therefore, we synthesized  $\alpha\text{-MnO}_2$  nanowires *via* a simple one-step hydrothermal method. Fig. S19† shows the typical XRD pattern of  $\alpha\text{-MnO}_2$ . Specifically, six diffraction peaks located at  $12.6$ ,  $18.0$ ,  $28.5$ ,  $37.4$ ,  $41.8$ , and  $49.7^\circ$  can be indexed to the (110), (200), (310), (211), (301), and (411) planes of  $\alpha\text{-MnO}_2$  (JCPDS no. 44-0141).<sup>41</sup> The SEM image shows that the  $\alpha\text{-MnO}_2$  nanowires are uniformly covered on the surfaces of the CC (Fig. S20†). The morphology of  $\alpha\text{-MnO}_2$  is further investigated by TEM. It shows that the diameter of the nanowire is  $\sim 60 \text{ nm}$  (Fig. S21a†). In addition, the HRTEM image shows that the three interplanar spacings are  $0.7$ ,  $0.25$ , and  $0.24 \text{ nm}$ , respectively, which are the features assigned to the (110), (400), and (211) planes of  $\alpha\text{-MnO}_2$  (Fig. S21b†). SAED (Fig. S19b,† the inset) ascertains that the prepared  $\alpha\text{-MnO}_2$  nanowire is single crystals. The element mapping (Fig. S21c†) shows that both Mn and O are concentrated in the entire sample, further confirming the composition of  $\alpha\text{-MnO}_2$  nanowires.

Before assembling the A-HSCs, we obtained tunnel  $\alpha\text{-MnO}_2$  through a one-step hydrothermal process, which will serve as

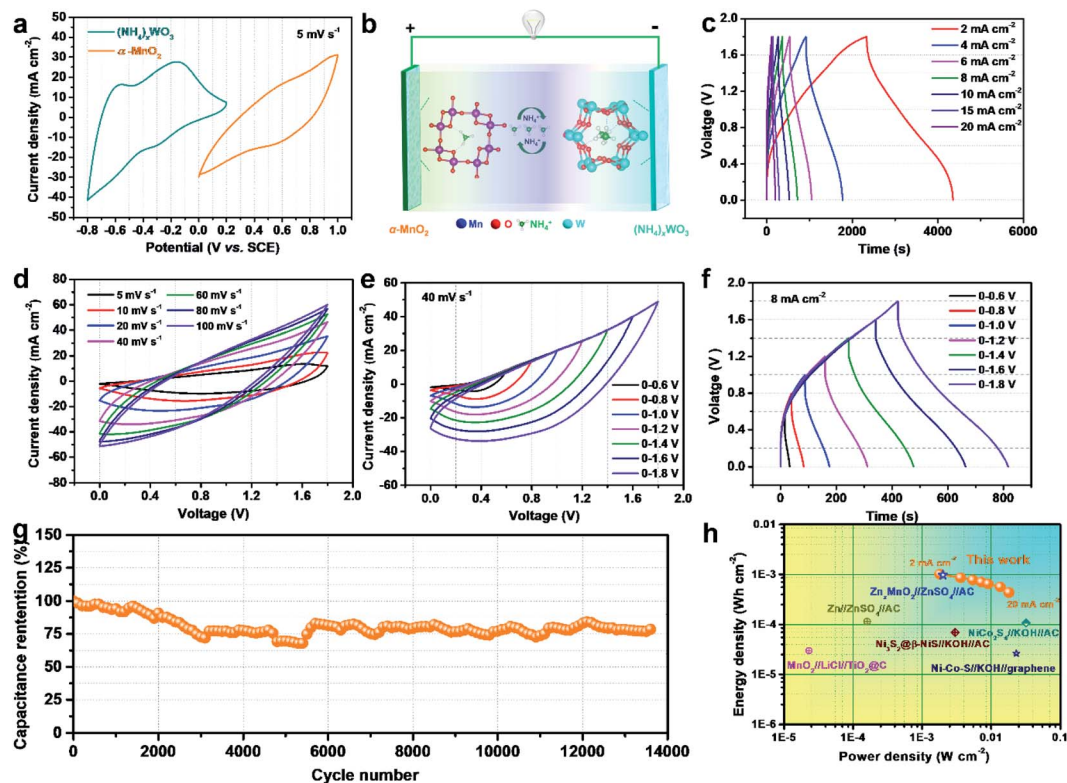


Fig. 4 Electrochemical performance of A-HSCs. (a) CV curves of the  $(\text{NH}_4)_x\text{WO}_3$  anode and the  $\alpha\text{-MnO}_2$  cathode at a scan rate of  $5 \text{ mV s}^{-1}$ . (b) Schematic illustration of aqueous  $(\text{NH}_4)_x\text{WO}_3/\alpha\text{-MnO}_2$  A-HSCs. (c) GCD and (d) CV curves were collected at different current densities and scan rates of the  $(\text{NH}_4)_x\text{WO}_3/\alpha\text{-MnO}_2$  A-HSCs. (e) CV (at  $40 \text{ mV s}^{-1}$ ) and (f) GCD (at  $8 \text{ mA cm}^{-2}$ ) curves of the  $(\text{NH}_4)_x\text{WO}_3/\alpha\text{-MnO}_2$  A-HSCs collected at different potential windows. (g) Cycling performance (at  $20 \text{ mA cm}^{-2}$ ), and (h) Ragone plots of the  $(\text{NH}_4)_x\text{WO}_3/\alpha\text{-MnO}_2$  A-HSCs.

the cathode of the A-HSC. Fig. 4a shows the CV curves of the  $\alpha\text{-MnO}_2$  nanowire and  $(\text{NH}_4)_x\text{WO}_3$  nanorod electrodes obtained in the three-electrode system, which exhibit the excellent capacitance and typical pseudo-capacitance performance of the  $\alpha\text{-MnO}_2$  cathode (Fig. S22<sup>†</sup>) and  $(\text{NH}_4)_x\text{WO}_3$  anode in the A-HSC, respectively. For the assembly of A-HSCs, tunnel-structured oxides are used as anode and cathode electrodes (Fig. 4b). Among them,  $(\text{NH}_4)_x\text{WO}_3$  is used as the anode,  $\alpha\text{-MnO}_2$  acts as the cathode, and the electrolyte is  $2.0 \text{ M NH}_4\text{Ac}$ . Fig. 4c and d show the GCD curves at different current densities and the CV curves at various scan rates of the A-HSC, respectively.  $(\text{NH}_4)_x\text{WO}_3/\alpha\text{-MnO}_2$  A-HSCs exhibit a quasi-triangular GCD profile and a rectangular CV similar to a traditional supercapacitor. The calculation results of GCD show that the areal capacitance of A-HSCs reaches  $2239.7 \text{ mF cm}^{-2}$  at  $2 \text{ mA cm}^{-2}$ , which is better than those of most reported supercapacitors (Table S3<sup>†</sup>), such as  $\text{Zn}_x\text{MnO}_2/\text{ACC Zn-HSCs}$  ( $1745.8 \text{ mF cm}^{-2}$  at  $2 \text{ mA cm}^{-2}$ ),<sup>41</sup>  $\text{AC//Zn foil Zn-HSCs}$  ( $1297 \text{ mF cm}^{-2}$  at  $0.16 \text{ mA cm}^{-2}$ ),<sup>46</sup> carbon nanotube//Zn nanoflake Zn-HSCs ( $83.2 \text{ mF cm}^{-2}$  at  $1 \text{ mA cm}^{-2}$ ),<sup>47</sup>  $\text{AC//Zn Zn-HSCs}$  ( $217.8 \text{ mF cm}^{-2}$  at  $0.08 \text{ mA cm}^{-2}$ ),<sup>48</sup>  $\text{GaN/MnO}_2/\text{MnON//G/M/M HSCs}$  ( $1915.5 \text{ mF cm}^{-2}$  at  $0.1 \text{ mA cm}^{-2}$ ),<sup>49</sup> and  $\text{CoHCF//AC HSCs}$  ( $351 \text{ mF cm}^{-2}$  at  $4.5 \text{ mA cm}^{-2}$ ).<sup>50</sup>

To find a suitable electrochemical window for the  $(\text{NH}_4)_x\text{WO}_3/\alpha\text{-MnO}_2$  A-HSC, the voltage windows of the CV curve (Fig. 4e) and GCD profiles (Fig. 4f) were adjusted from 0.6

to 1.8 V, which confirmed that the voltage of the assembled A-HSC can be stabilized to 1.8 V. The cycling performance of  $(\text{NH}_4)_x\text{WO}_3/\alpha\text{-MnO}_2$  A-HSCs in  $2.0 \text{ NH}_4\text{Ac}$  electrolyte is shown in Fig. 4g. The A-HSC displays outstanding cycle performance, with a capacity retention rate of 78.6% after 13 600 cycles at  $20 \text{ mA cm}^{-2}$ . Fig. 4h further shows the areal power and energy density of  $(\text{NH}_4)_x\text{WO}_3/\alpha\text{-MnO}_2$  A-HSCs, where A-HSCs can provide a maximum areal energy density of  $1010.1 \mu\text{Wh cm}^{-2}$ , and a peak areal power density of  $18.0 \text{ mW cm}^{-2}$ . Importantly, the areal energy density of the as-assembled A-HSCs is much higher than those of many reported supercapacitors (Table S3<sup>†</sup>), such as  $\text{Zn}_x\text{MnO}_2/\text{ACC Zn-HSCs}$  ( $969.9 \mu\text{Wh cm}^{-2}$ ),<sup>41</sup>  $\text{AC//Zn foil Zn-HSCs}$  ( $115.4 \mu\text{Wh cm}^{-2}$ ),<sup>46</sup> carbon nanotube//Zn nanoflake Zn-HSCs ( $29.6 \mu\text{Wh cm}^{-2}$ ),<sup>47</sup>  $\text{AC//Zn Zn-HSCs}$  ( $67.2 \mu\text{Wh cm}^{-2}$ ),<sup>48</sup>  $\text{GaN/MnO}_2/\text{MnON//G/M/M HSCs}$  ( $61 \mu\text{Wh cm}^{-2}$ ),<sup>49</sup> and  $\text{CoHCF//AC HSCs}$  ( $191.25 \mu\text{Wh cm}^{-2}$ ).<sup>50</sup> The above results show the areal energy density, power density, and long cycling life of  $(\text{NH}_4)_x\text{WO}_3/\alpha\text{-MnO}_2$  A-HSCs, which can perfectly meet the frenetic demand for high-performance energy storage.

To reveal the intrinsic  $\text{NH}_4^+$  storage mechanism of the A-HSC, *ex situ* XRD and XPS characterization are performed. The first GCD profile of the  $(\text{NH}_4)_x\text{WO}_3$  anode is shown in Fig. 5a. To investigate the evolution of the anode crystal structure during A-HSC charge and discharge, the corresponding *ex situ* XRD pattern is shown in Fig. 5b. The pristine sample showed typical characteristic diffraction peaks of  $(\text{NH}_4)_x\text{WO}_3$ . Moreover,

the characteristic diffraction remained almost unchanged during the charging and discharging process, indicating that the  $\text{NH}_4^+$  intercalation/de-intercalation process is reversible, which is beneficial to the cycling performance of A-HSCs. Fig. 5c shows an enlarged part of the red rectangle marked in Fig. 5b. The characteristic peaks corresponding to the (002), (102), and (200) planes of  $(\text{NH}_4)_x\text{WO}_3$  have undergone reversible shifts during the charging/discharging process. Specifically, the characteristic peak of the discharge process shifts slightly to a low angle and shifts to a high angle after charging. This indicates that the intercalation/de-intercalation process of  $\text{NH}_4^+$  is reversible and that there is a strong bond between the intercalation of  $\text{NH}_4^+$  and the tunnel structure of the host. *Ex situ* XPS spectra are collected to detect the chemical states of N and W in the charged/discharged state of  $(\text{NH}_4)_x\text{WO}_3$ . As shown in Fig. 5d, the characteristic peaks of N 1s at 400.5 and 402.3 eV can be attributed to N-W and N-NH<sub>2</sub>, respectively.<sup>36</sup> We define the ratio of the characteristic peak areas of N-NH<sub>2</sub> to N-W as the parameter  $x$ . The value of  $x$  decreases gradually with the charging process and increases gradually with the discharge, indicating that  $\text{NH}_4^+$  is reversible in the process of charging and discharging. In addition, the ratio of the characteristic peak

areas of W-N to W-O in the W 4f spectra has a phenomenon like that in the N 1s spectra (Fig. 5e).

To further reveal the bonding mechanism of the  $\text{NH}_4^+$  storage of the A-HSC, the *ex situ* Fourier transform infrared (FT-IR, Fig. 5f) spectrum of the  $(\text{NH}_4)_x\text{WO}_3$  anode at the charge and discharge states was collected. The W-O-W characteristic vibration of  $(\text{NH}_4)_x\text{WO}_3$  is shown in the high-frequency region ( $\sim 965\text{ cm}^{-1}$ ).<sup>51</sup> The region from  $1400\text{--}1800\text{ cm}^{-1}$  is generally the scissoring vibration of the N-H and O-H bonds, and the peaks close to  $1747\text{ cm}^{-1}$  are assigned to H-bonded N-H stretch.<sup>20</sup> Furthermore, the low frequency ( $2855\text{ cm}^{-1}$ ) is attributed to the stretching of N-H bonded to the WO host, and  $2926\text{ cm}^{-1}$  is the stretching of non-bonded N-H.<sup>52</sup> Specifically, the initial sample also showed a relatively weak signal due to the pre-intercalation of  $\text{NH}_4^+$ . In a typical charging/discharging process, the signal after discharge is enhanced and then weakened after charging, which indicates that the formation/breaking of hydrogen bonds is related to the intercalation/de-intercalation of  $\text{NH}_4^+$  in the  $(\text{NH}_4)_x\text{WO}_3$  host. The above *ex situ* characterization clearly shows that  $\text{NH}_4^+$  is reversibly intercalated/de-intercalated in the tunnel structured  $(\text{NH}_4)_x\text{WO}_3$  during discharge/charging.

Based on the above results, a detailed  $\text{NH}_4^+$  storage mechanism of  $(\text{NH}_4)_x\text{WO}_3$  is proposed in Fig. 5g.  $\text{NH}_4^+$  pre-

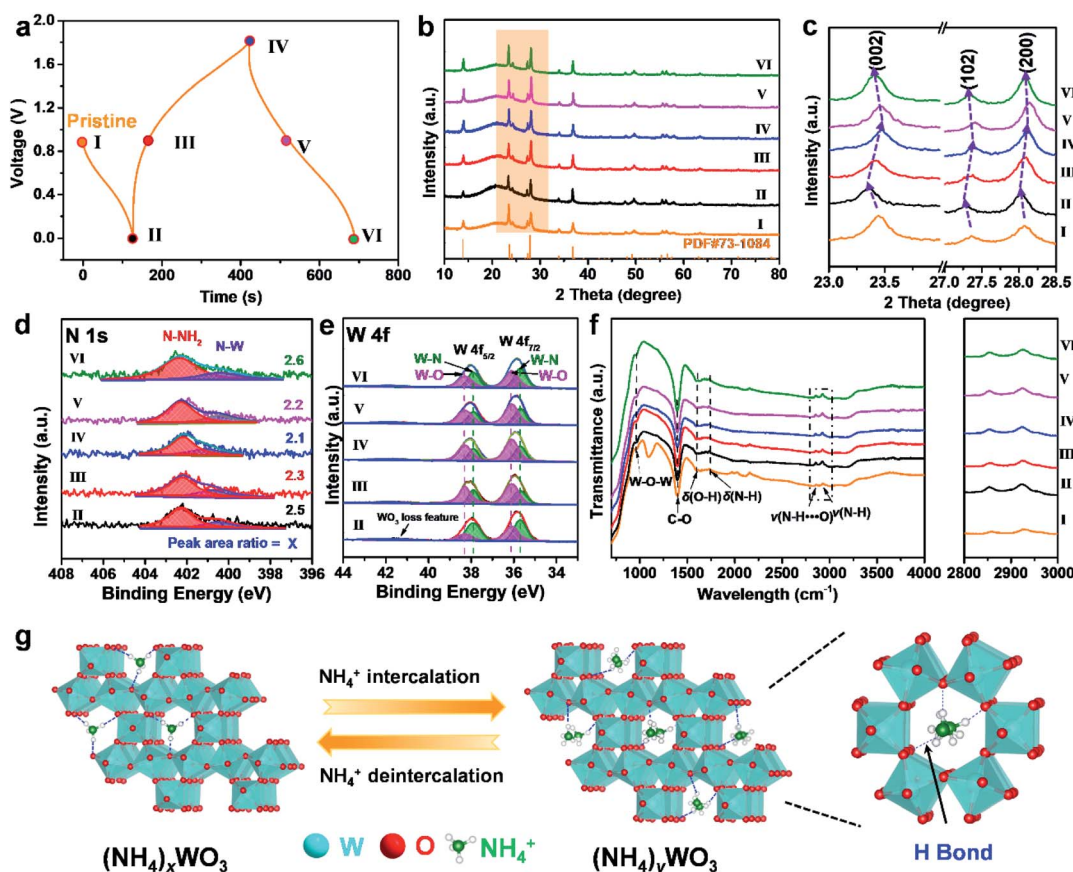


Fig. 5 The  $\text{NH}_4^+$  storage mechanism of the  $(\text{NH}_4)_x\text{WO}_3$  anode. (a) GCD curves of  $(\text{NH}_4)_x\text{WO}_3//\alpha\text{-MnO}_2$  A-HSCs at  $8\text{ mA cm}^{-2}$ . (b) Corresponding *ex situ* XRD patterns and (c) the enlarged section of the marked-out red rectangle. (d) Core-level N 1s and (e) W 4f XPS spectra, (f) FT-IR spectrum and magnification of the dashed region of the  $(\text{NH}_4)_x\text{WO}_3$  anode at different charge/discharge states. (g) Schematic illustration of the  $\text{NH}_4^+$  storage mechanism in the  $(\text{NH}_4)_x\text{WO}_3$  anode.

intercalation in  $\text{WO}_3$  can stabilize the tunnel structure and increase the adsorption energy of  $\text{NH}_4^+$ . During the charge/discharge process,  $\text{NH}_4^+$  ions interact strongly in the  $(\text{NH}_4)_x\text{WO}_3$  host structure through hydrogen bonds with the adjacent WO tunnel structure, further stabilizing  $(\text{NH}_4)_x\text{WO}_3$  and contributing to the subsequent highly reversible  $\text{NH}_4^+$  intercalation/de-intercalation.

## Conclusions

In summary, we have demonstrated the relationship between the hydrogen bond and pseudocapacitance behavior of the tunnel structure during  $\text{NH}_4^+$  storage, and successfully constructed a novel A-HSC. Ammonium ion pre-intercalation can stabilize and improve the  $\text{NH}_4^+$  storage performance of the tunnel structured  $h\text{-WO}_3$ . Moreover, the tunnel structured  $\alpha\text{-MnO}_2$  cathode has been proved to have a fast  $\text{NH}_4^+$  storage mechanism similar to that of the  $(\text{NH}_4)_x\text{WO}_3$  anode. The newly developed  $(\text{NH}_4)_x\text{WO}_3/\alpha\text{-MnO}_2$  A-HSC delivers an ultra-high areal capacitance of  $2239.7\text{ mF cm}^{-2}$  and a peak areal energy density of  $1010.1\text{ }\mu\text{Wh cm}^{-2}$ . In addition, first-principles electronic structure calculations show that the charge accumulates in the tunnel structure transfer orientation bond, and the pre-inserted  $\text{NH}_4^+$  is beneficial in lowering the storage energy barrier of  $\text{NH}_4^+$  storage and achieve fast ion diffusion. This work establishes the paradigm of a non-metal ion stabilized tunnel structure for fast  $\text{NH}_4^+$  storage, providing a new perspective for the next generation of electrochemical energy storage devices.

## Author contributions

Q. C. and Y. T. conceived and planned the experiments. M. S. and X. Z. prepared the samples and carried out the experiments. G. H. conducted TEM measurement of the samples. M. S. and J. Z. carried out the DFT calculations. Q. C., M. S., and Y. T. interpreted all the results and wrote and edited the manuscript. All authors have reviewed, discussed and approved the results and conclusions of this article.

## Conflicts of interest

There are no conflicts to declare.

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## Notes and references

- 1 Y. Liang, J. Yan, S. Gheyhani, K. Y. Lee, P. Liu, A. Facchetti and Y. Yao, *Nat. Mater.*, 2017, **16**, 841–848.
- 2 D. Kundu, B. D. Adams, V. Duffort, S. H. Vajargah and L. F. Nazar, *Nat. Energy*, 2016, **1**, 16119.
- 3 H. Li, C. J. Firby and A. Y. Elezzabi, *Joule*, 2019, **3**, 2268–2278.
- 4 P. Poizot, S. Laruelle, S. Grugeon, L. Dupont and J.-M. Tarascon and, *Nature*, 2000, **407**, 496–499.
- 5 Z. Wang, L. Zhou and X. Lou, *Adv. Mater.*, 2012, **43**, 1903–1911.
- 6 V. Palomares, P. Serras, I. Villaluenga, K. B. Hueso, J. Carretero-González and T. Rojo, *Energy Environ. Sci.*, 2012, **5**, 5884–5901.
- 7 N. Yabuuchi, K. Kubota, M. Dahbi and S. Komaba, *Chem. Rev.*, 2014, **114**, 11636–11682.
- 8 D. Su, A. McDonagh, S. Z. Qiao and G. Wang, *Adv. Mater.*, 2017, **29**, 1604007.
- 9 F. Wang, O. Borodin, T. Gao, X. Fan, W. Sun, F. Han, A. Faraone, J. A. Dura, K. Xu and C. Wang, *Nat. Mater.*, 2018, **17**, 543–549.
- 10 Y. Xu, X. Deng, Q. Li, G. Zhang, F. Xiong, S. Tan, Q. Wei, J. Lu, J. Li, Q. An and L. Mai, *Chem*, 2019, **5**, 1194–1209.
- 11 M. Wang, C. Jiang, S. Zhang, X. Song, Y. Tang and H. M. Cheng, *Nat. Chem.*, 2018, **10**, 667–672.
- 12 S. Wang, S. Huang, M. Yao, Y. Zhang and Z. Niu, *Angew. Chem., Int. Ed.*, 2020, **59**, 11800–11807.
- 13 T. Sun, H. Du, S. Zheng, J. Shi and Z. Tao, *Adv. Funct. Mater.*, 2021, **31**, 2010127.
- 14 Q. Chen, J. Jin, M. Song, X. Zhang, H. Li, J. Zhang, G. Hou, Y. Tang, L. Mai and L. Zhou, *Adv. Mater.*, 2022, **34**, 2107992.
- 15 J. W. Erisman, M. A. Sutton, J. Galloway, Z. Klimont and W. Winiwarter, *Nat. Geosci.*, 2008, **1**, 636–639.
- 16 G. Marnellos and M. Stoukides, *Science*, 1998, **282**, 98–100.
- 17 G. Liang, F. Mo, X. Ji and C. Zhi, *Nat. Rev. Mater.*, 2020, **6**, 109–123.
- 18 X. Wu, Y. Qi, J. J. Hong, Z. Li, A. S. Hernandez and X. Ji, *Angew. Chem., Int. Ed.*, 2017, **56**, 13026–13030.
- 19 D. Chao and H. J. Fan, *Chem*, 2019, **5**, 1357–1360.
- 20 Y. Song, Q. Pan, H. Lv, D. Yang, Z. Qin, M.-Y. Zhang, X. Sun and X.-X. Liu, *Angew. Chem., Int. Ed.*, 2021, **60**, 5718–5722.
- 21 S. Dong, W. Shin, H. Jiang, X. Wu, Z. Li, J. Holoubek, W. F. Stickle, B. Key, C. Liu, J. Lu, P. A. Greaney, X. Zhang and X. Ji, *Chem*, 2019, **5**, 1537–1551.
- 22 Y. Ye, H. H. Ngo, W. Guo, Y. Liu, S. W. Chang, D. D. Nguyen, H. Liang and J. Wang, *Bioresour. Technol.*, 2018, **268**, 749–758.
- 23 X. Wang, C. Bommier, Z. Jian, Z. Li, R. S. Chandrabose, I. A. Rodríguez-Pérez, P. A. Greaney and X. Ji, *Angew. Chem., Int. Ed.*, 2017, **56**, 2909–2913.
- 24 X. Wu, Y. Xu, H. Jiang, Z. Wei, J. J. Hong, A. S. Hernandez, F. Du and X. Ji, *ACS Appl. Energy Mater.*, 2018, **1**, 3077–3083.
- 25 M. Xia, X. Zhang, H. Yu, Z. Yang, S. Chen, L. Zhang, M. Shui, Y. Xie and J. Shu, *Chem. Eng. J.*, 2021, **421**, 127759.
- 26 C. Li, W. Yan, S. Liang, P. Wang, J. Wang, L. Fu, Y. Zhu, Y. Chen, Y. Wu and W. Huang, *Nanoscale Horiz.*, 2019, **4**, 991–998.
- 27 X. Zhang, M. Xia, T. Liu, N. Peng, H. Yu, R. Zheng, L. Zhang, M. Shui and J. Shu, *Chem. Eng. J.*, 2021, **421**, 127767.
- 28 M. R. Lukatskaya, O. Mashtalir, C. E. Ren, Y. Dall'Agnese, P. Rozier, P. L. Taberna, M. Naguib, P. Simon, M. W. Barsoum and Y. Gogotsi, *Science*, 2013, **341**, 1502–1505.

- 29 H. Li, J. Yang, J. Cheng, T. He and B. Wang, *Nano Energy*, 2020, **68**, 104369.
- 30 G. Liang, Y. Wang, Z. Huang, F. Mo, X. Li, Q. Yang, D. Wang, H. Li, S. Chen and C. Zhi, *Adv. Mater.*, 2020, **32**, 1907802.
- 31 Y. Z. Zhang, J. Liang, Z. Huang, Q. Wang, G. Zhu, S. Dong, H. Liang and X. Dong, *Adv. Sci.*, 2022, **9**, 2105158.
- 32 J. Jin, X. Geng, Q. Chen and T. L. Ren, *Nano Micro Lett.*, 2022, **14**, 64.
- 33 P. G. Dickens and D. J. Neild, *J. Solid State Chem.*, 1973, **7**, 474–476.
- 34 Z. Yi, C. Gang, Y. Yu, Y. Zhou and H. Fang, *Appl. Surf. Sci.*, 2015, **362**, 182–190.
- 35 K. Juntaracena, T. Yuangkaew, M. Horprathum, N. Triroj and P. Jaroenapibal, *Vib. Spectrosc.*, 2021, **115**, 103276.
- 36 H. Yan, C. Tian, L. Sun, B. Wang, L. Wang, J. Yin, A. Wu and H. Fu, *Energy Environ. Sci.*, 2014, **7**, 1939–1949.
- 37 A. Fularz, S. Almohammed and J. H. Rice, *J. Phys. Chem. C*, 2020, **124**, 25351–25360.
- 38 J. Li, S. Chen, N. Yang, M. Deng, S. Ibraheem, J. Deng, J. Li, L. Li and Z. Wei, *Angew. Chem., Int. Ed.*, 2019, **58**, 7035–7039.
- 39 N. Han, K. R. Yang, Z. Lu, Y. Li and X. Sun, *Nat. Commun.*, 2018, **9**, 924.
- 40 N. Jabeen, A. Hussain, Q. Xia, S. Sun, J. Zhu and H. Xia, *Adv. Mater.*, 2017, **29**, 1700804.
- 41 Q. Chen, J. Jin, Z. Kou, C. Liao, Z. Liu, L. Zhou, J. Wang and L. Mai, *Small*, 2020, **16**, 202000091.
- 42 Z. Teng, L. Wan, S. Sun, C. Qi, S. Jiao, Q. Xia and X. Hui, *Adv. Mater.*, 2017, **29**, 1604167.
- 43 Q. Chen, J. Jin, Z. Kou, J. Jiang, Y. Fu, Z. Liu, L. Zhou and L. Mai, *J. Mater. Chem. A*, 2020, **8**, 13114–13120.
- 44 S. Liu, Y. Zeng, M. Zhang, S. Xie, Y. Tong, F. Cheng and X. Lu, *J. Mater. Chem. A*, 2017, **5**, 21460–21466.
- 45 R. Wang, Y. Lu, L. Zhou, Y. Han, J. Ye, W. Xu and X. Lu, *Electrochim. Acta*, 2018, **283**, 639–645.
- 46 P. Zhang, Y. Li, G. Wang, F. Wang, S. Yang, F. Zhu, X. Zhuang, O. G. Schmidt and X. Feng, *Adv. Mater.*, 2019, **31**, 1806005.
- 47 G. Sun, H. Yang, G. Zhang, J. Gao, X. Jin, Y. Zhao, L. Jiang and L. Qu, *Energy Environ. Sci.*, 2018, **11**, 3367–3374.
- 48 L. Dong, X. Ma, Y. Li, L. Zhao, W. Liu, J. Cheng, C. Xu, B. Li, Q.-H. Yang and F. Kang, *Energy Storage Mater.*, 2018, **13**, 96–102.
- 49 S. Wang, Y. Shao, W. Liu, Y. Wu and X. Hao, *J. Mater. Chem. A*, 2018, **6**, 13215–13224.
- 50 J.-G. Wang, Z. Zhang, X. Zhang, X. Yin, X. Li, X. Liu, F. Kang and B. Wei, *Nano Energy*, 2017, **39**, 647–653.
- 51 J. Shi, Z. Cheng, L. Gao, Y. Zhang, J. Xu and H. Zhao, *Sens. Actuators, B*, 2016, **230**, 736–745.
- 52 A. Zecchina, L. Marchese, S. Bordiga, C. Pazè and E. Gianotti, *J. Phys. Chem. B*, 1997, **101**, 10128–10135.